

Part A · Multiple Choice

2 points each

1. **c) how well a surface reflects light**

p. 83 — a surface with high albedo reflects a large percentage of the light that hits it. A snowfield is the book's example.

2. **d) changes state**

p. 84 — and while a substance is changing state its temperature does not change; all the energy goes into the change of state itself.

3. **a) it takes a great deal of energy to change the temperature of water**

p. 85 — this is why a puddle stays cooler than an asphalt parking lot receiving the very same sunlight, and why oceans moderate coastal temperatures.

4. **a) 0° and 30° latitude**

p. 91 — the Ferrel cell runs from about 30° to 50–60°, and the Polar cell from about 50–60° to the pole.

5. **b) the trade winds**

p. 92 — named because in the age of sailing ships they were reliable for trade. The deflection to the right is the Coriolis effect.

6. **b) air that rose, cooled, and dropped its precipitation on the windward side then descends and warms**

p. 89 — the same descending, drying air produces the rain shadow effect, which the text credits with creating many of the world's deserts.

Part B · True or False

1 point each

7. **FALSE**

They are named for the direction the winds come from. The westerlies blow from the west toward the east. (p. 93)

8. **TRUE**

p. 93 — a jet stream is a fast-flowing river of air at the boundary between troposphere and stratosphere. The polar jet stream is the world's most powerful because the temperature contrast at the polar front is the sharpest.

Part C · Short Answer

3 points each

9. **Temperature measures how fast a material's atoms are vibrating; heat measures the material's total energy. The candle flame has the higher temperature; the bathtub contains far more heat.**

Rubric: 1 pt temperature = speed of atomic vibration. 1 pt heat = total energy of the material. 1 pt for the comparison: the flame is hotter but its hot region is very small, while the bathtub is cooler but holds vastly more vibrating atoms and therefore greater total energy. (pp. 84–85)

10. **Hadley cell (0°–30°), Ferrel cell (30° to 50–60°), and Polar cell (50–60° to the pole).**

Rubric: 1 pt per cell, requiring both the name and a roughly correct range. Accept 60° as the boundary between Ferrel and Polar. Bonus context, not required for credit: adjacent cells share their rising and descending limbs, the Southern Hemisphere mirrors these three, and the junction of the Ferrel and Polar cells is the polar front, where the polar jet stream is found. (pp. 91–93)